Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Regular Examination December 2024 (2024 Scheme)

Course Code: GAMAT101

Course Name: MATHEMATICS FOR INFORMATION SCIENCE-1

Max. Marks: 60 Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks)

		CO	Marks
1	At what points are the function $g(x) = \begin{cases} \frac{x^2-x-6}{x-3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$ continuous.	CO1	3
2	Find the first and second derivatives of $\frac{x^3+7}{x}$.	CO1	3
3	Find $\lim_{(x,y)\to(0,0)} \frac{1-\cos(xy)}{xy}$	CO2	3
4	Show that $w_{xy} = w_{yx}$ where $w = e^x + x\,lny + y\,lnx$.	CO2	3
5	What are the directions of zero change in $f(x,y) = \frac{x^2}{2} + \frac{y^2}{2}$ at (1,1)	CO3	3
6	Find the critical points of the function $f(x,y) = 5xy - 7x^2 + 3x - 6y + 2$.	CO3	3
7	Find the extreme points of $f(x,y) = xy$ subject to the constraints $x + y = 16$	CO4	3
8	Form the LPP by maximising the profit: DC Drug company produces two types of liquid pain killer, N(normal) and S(super). Each bottle of N requires 2 units of drug A, 1 unit of drug B and 1 unit of drug C. Each bottle of S requires 1 unit of A, 1 unit of B, 3 units of C. The company is able to produce each week only 1400 units of A, 800 units of B and 1800 units of C. The profit per bottle of N and S is \$11 and \$15 respectively.	CO4	3

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PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

9	a)	Find $\lim_{x\to 0} \frac{\frac{1}{x-1}+\frac{1}{x+1}}{x}$	CO1	3
	b)	Find the linearization of $f(x) = cosx$ at $x = \frac{\pi}{2}$.	CO1	3
	c)	Determine the concavity of $f(x) = x^3 - 6x^2 + 9x + 1$.	CO1	3
10	a)	Show that $y = \|x\|$ is differentiable on $(-\infty, 0)$ and $(0, \infty)$, but has no derivative at $x = 0$.	CO1	6
	b)	The area A of a circle is related to its diameter D by the equation $A = \frac{\pi}{4}D^2$. How fast does the area change with respect to the diameter when diameter is 10m.	CO1	3

Module -2

11	a)	Find $\frac{\partial^2 w}{\partial y \partial x}$ if $w = xy + \frac{e^y}{y^2+1}$	CO2	3
	b)	Find the limit of the function $\lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}}$.	CO2	3
	c)	Find $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$ for the function $f(x,y,z) = e^{\frac{x}{y}} + e^{\frac{z}{y}}$	CO2	3
12	a)	Find the domain and range of the function $g(x,y) = \ln(x^2+y^2)$	CO2	3
	b)	If $w = e^{x^2 y}, x = \sqrt{uv}, y = \frac{1}{v}$, find $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ at (2,2) using chain rule.	CO2	6

Module -3

13	a)	If $w = f(x-y, y-z, z-x)$, show that $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$.	CO3	5
	b)	Find the directional derivative at (1,-1) of $g(x,y) = \frac{x-y}{xy+2}$ in the direction of $\vec{v} = 12\vec{i} + 5\vec{j}$.	CO3	4
14	a)	Find the local extrema of $f(x,y) = 10xye^{-(x^2+y^2)}$	CO3	6

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	b)	Find the gradient of $f(x,y) = x^2 - xy + y^2 - y$ at $(1,-1)$	CO3	3

Module -4

15	a)	Find the point $P(x,y,z)$ on the plane $2x + y - z - 5 = 0$ that is close to the origin.	CO4	6
	b)	Minimise the function $f(x,y) = x^2 + y^2$ with starting point (1,1) with step size, $\alpha = 0.1$ in two iterations by the method of steepest descent.	CO4	3
16	a)	Find the largest and smallest values the function $f(x,y) = xy$ takes on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$ by Lagrange's multiplier method.	CO4	5
	b)	Solve the following LPP graphically: Maximise $z = 6x_1 + 11\,x_2$ subject to $2x_1 + x_2 \leq 104$ $x_1 + 2x_2 \leq 76$ $x_1 \geq 0, x_2 \geq 0$	CO4	4

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